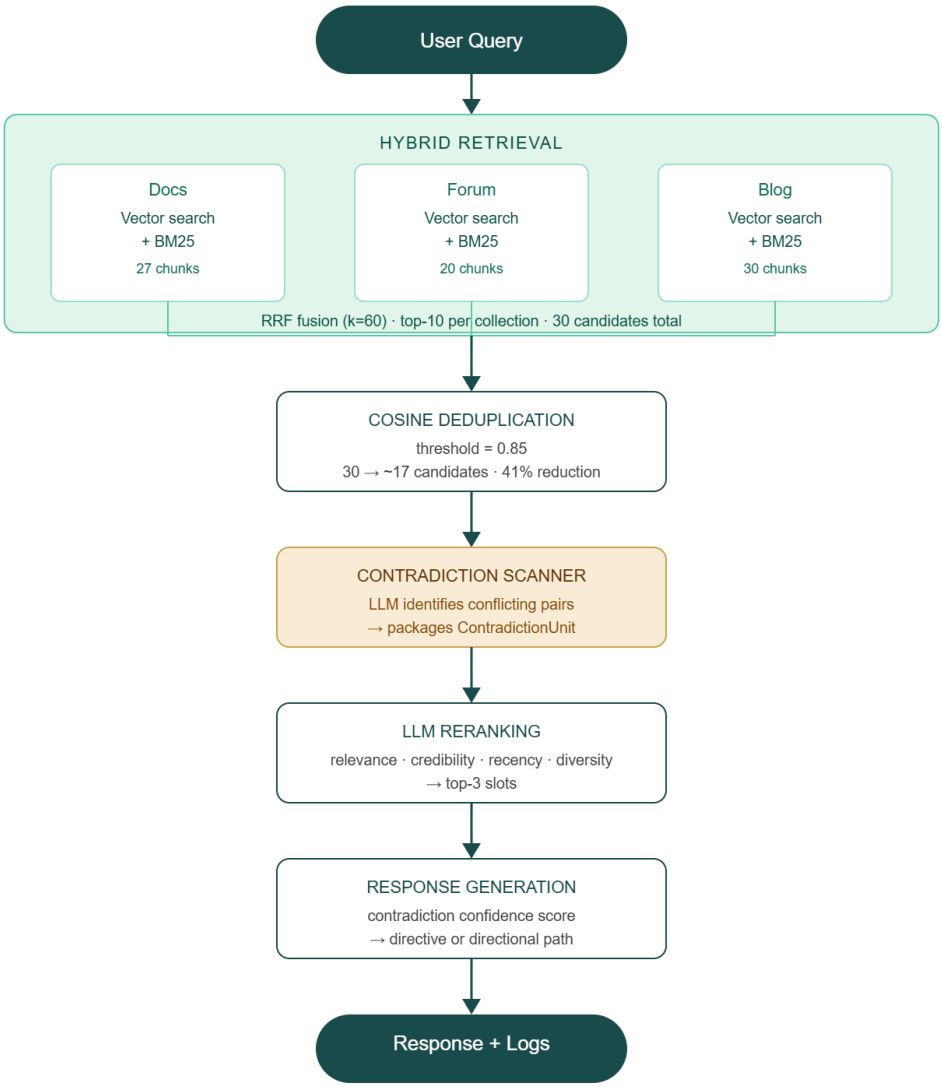
FertiliQ Multi-Source RAG System

Github repo: https://github.com/yz3447-bot/fertiliq-rag.git
Video walkthrough: FertiliQ video walkthrough 2026-04-19 10-54-44.mp4

1. System Overview

FertiliQ is a fictional preconception health platform that combines genetic testing, biomarker tracking, and AI-powered advisory services. The platform ingests three distinct categories of knowledge, clinical documentation, practitioner-moderated forums, and evidence-based blog content, and surfaces synthesized, actionable answers to user questions about fertility, supplementation, and reproductive health. The system is designed to handle a common challenge in health-tech RAG: sources that contradict one another due to evolving research, outdated documentation, or differing levels of clinical authority.

Pipeline Architecture



TECH STACK	
Embedding	gemini-embedding-001 via google-genai SDK
Generation	gemini-2.5-flash
Vector store	ChromaDB PersistentClient · cosine space · 3 collections
Lexical search	BM25Okapi rank-bm25
UI	Streamlit custom CSS · DM Serif Display · DM Sans
Logging	JSON append log logs/query_log.json
Language	Python 3.11

2. Data Creation and Chunking Strategy

Why FertiliQ

FertiliQ was chosen because preconception health sits at the intersection of rapidly evolving clinical research and high-stakes personal decisions. This creates exactly the conditions where a RAG system is most stressed: documentation gets superseded, forum advice reflects outdated practice, and users need clear directives rather than hedged non-answers. The same structural challenges appear in real-world medical RAG systems covering oncology protocols, drug interaction databases, and chronic disease management guidelines, making FertiliQ a realistic and instructive proxy.

Documentation: Header-Based Parent-Child Chunking

Clinical documentation tends to have hierarchical structure, a parent section establishes context (e.g., "MTHFR Variants and Folate Metabolism") while child sections contain specific actionable content (e.g., dosing tables, contraindications). Splitting only at the child level risks losing the interpretive context provided by the parent. The system therefore stores full parent sections as a lookup table and injects parent content into child chunk metadata at retrieval time. Child chunks are embedded and stored in ChromaDB; the parent content is surfaced only when a child chunk is retrieved, expanding the context window passed to the LLM without inflating the index.

Forum: QA-Pair Chunking

Forum posts have a natural semantic unit: the question and its accepted answer together represent a complete, grounded exchange. Chunking questions and answers separately produces fragments that lose

their meaning in isolation. By treating each QA pair as a single chunk, the system preserves the reasoning chain from user concern to expert response. Forum chunks additionally carry vote count and acceptance status as metadata, which feeds directly into the credibility scoring used during reranking.

Blog: Semantic Paragraph Chunking

Blog articles are narrative and exploratory rather than structured. They do not have clean section headers that would support hierarchical chunking, nor do they have the call-and-response structure of forum posts. Semantic paragraph chunking, where each chunk corresponds to a coherent paragraph or two, preserves the argumentative flow of the writing while keeping chunk length within embedding model limits. This approach also reduces the risk of splitting a key claim from its supporting evidence, which would degrade retrieval precision.

Deliberate Contradictions

Two specific contradictions were embedded to test the pipeline's ability to detect and resolve conflicting information:

MTHFR and folic acid dosing. A legacy forum post from August 2022 recommends 400 mcg of standard folic acid daily for MTHFR C677T carriers. A 2024 clinical documentation update (v2.1) recommends switching entirely to active methylfolate (5-MTHF) at 800 mcg, citing evidence that synthetic folic acid is ineffective and potentially harmful for individuals with impaired MTHFR enzyme activity. This temporal contradiction is clinically significant: following the older advice is contraindicated by current evidence.

CoQ10 dosing for egg quality. A 2022 blog post recommends 200 mg per day. A 2024 documentation update increases the recommendation to 400 to 600 mg per day for women over 35, citing newer RCT data on mitochondrial function in oocytes. This version-type contradiction tests whether the system can identify superseded dosing guidance and surface the more authoritative, more recent recommendation.

3. Retrieval System Design

Hybrid Retrieval

Each of the three ChromaDB collections is searched independently using two methods. Vector search retrieves semantically similar chunks using Gemini embeddings. BM25 retrieves chunks with high lexical overlap with the query, which is particularly effective for queries containing specific clinical terms (gene variant names, supplement names, biomarker abbreviations) that may not have strong semantic neighbors in the embedding space.

The results from vector and BM25 searches within each collection are merged using Reciprocal Rank Fusion with k equal to 60, then the top 10 candidates per collection are retained, yielding 30 candidates in total.

Why RRF Over Manual Weight Tuning

Manual weight tuning (e.g., 0.7 * vector_score + 0.3 * bm25_score) requires a labeled evaluation set to calibrate and produces brittle scores that are not comparable across queries with different length distributions. RRF uses only rank positions, not raw scores, which makes the fusion stable across collections with very different score scales. The k parameter of 60 is a standard default that dampens the influence of top-ranked results without completely discounting lower-ranked ones. It has been shown empirically to perform well across a wide range of retrieval tasks without tuning.

Fixed Top-10 Per Collection

Retrieving 10 candidates per collection ensures that no single source type dominates the candidate set before reranking. Given three collections of unequal size (27 documentation chunks, 20 forum chunks, 30 blog chunks), a global top-30 without per-collection caps would consistently over-represent the largest collection. The fixed cap enforces source diversity at the retrieval stage, before any credibility or relevance judgements are made.

Source Credibility Hierarchy

Documentation carries a credibility score of 1.0, blogs 0.67, and forum posts 0.33. These weights are not used to filter at retrieval time but feed into the contradiction confidence calculation and LLM reranking prompt. The rationale is that clinical documentation is authored, reviewed, and versioned by medical professionals; blogs represent synthesized but non-peer-reviewed commentary; and forum posts reflect individual experience, however well-intentioned. This hierarchy mirrors how clinicians themselves weight information sources.

4. Reranking Mechanism

Single LLM Reranking Call

Rather than using separate passes for dynamic weight tuning and reranking, the system delegates the entire slot selection to a single LLM call. The reranking prompt presents all deduplicated candidates (and any contradiction units) to the model and asks it to select the three most valuable slots for answering the user's query. This approach reduces latency by eliminating a separate weighting pass and allows the model to reason holistically about relevance, credibility, recency, and diversity simultaneously.

What the LLM Considers

The reranking prompt instructs the model to prioritize: direct relevance to the query intent, source credibility (documentation over blog over forum), recency of information, and diversity across source types. The prompt explicitly flags contradiction units as needing special handling. If two chunks directly contradict each other, they should be included together as a single slot so the generation stage can address the conflict transparently.

Deduplication Before Reranking

Deduplication is performed before the LLM reranking call using cosine similarity on chunk embeddings. Any pair of chunks with cosine similarity above 0.85 is treated as a near-duplicate; the chunk with the lower RRF score is removed. This step is intentionally rule-based rather than LLM-based: deduplication is a mechanical task with a clear threshold criterion, and using an LLM for it would add latency and cost without meaningful quality benefit.

Across the 10 example queries logged, deduplication reduced the candidate pool from 30 to an average of 17.4 candidates, which is a 41% reduction. The most aggressive deduplication occurred on the biomarker sync query (30 to 20) and the follicular phase foods query (30 to 17). This reduction meaningfully shortens the context presented to the reranking and generation models.

5. Contradiction Handling

Three Contradiction Types

The system recognizes three categories of contradiction. Temporal contradictions arise when one source recommends something that a newer source explicitly supersedes, the MTHFR folic acid case is a clear example. Version contradictions occur when different document versions carry different guidance, the CoQ10 dosing update from v1.0 to v2.1 is representative. Credibility contradictions arise when a lower-credibility source such as a forum post makes a claim directly at odds with a higher-credibility source such as clinical documentation, without a clear temporal dimension.

Pre-Reranking Contradiction Scan

Contradiction detection runs on the deduplicated candidate pool, before reranking. This ordering matters: running it after reranking would mean the LLM has already selected a top-3 that may not include both sides of a contradiction. By detecting contradictions first, the system can package both conflicting chunks into a ContradictionUnit and present that unit to the reranker as a single slot, ensuring both sides reach the generation stage.

Contradiction Confidence Score

The confidence score is the mean of three components, each normalized to the range 0 to 1:

- Credibility gap: the absolute difference between the two sources' credibility scores
- Time gap (normalized): the number of years between publication dates, capped at 5 years (so a 5-year gap yields 1.0)
- Specificity: 0.9 if the contradiction description contains numeric values (doses, ranges, thresholds), 0.5 otherwise

A high confidence score (above 0.70) triggers a directive prompt that instructs the model to lead with the correct, current recommendation and explicitly explain why the older guidance is outdated. A low or moderate confidence score triggers a directional prompt that acknowledges the disagreement, offers a practical starting point, and avoids paralysing the user with uncertainty. In both cases, the system never deflects to a generic "consult a doctor" response as its primary output, and it always provides a concrete, actionable recommendation first.

Real Examples from the 10 Queries

Query 1 (MTHFR C677T, confidence 0.657): the system detected a temporal contradiction between the 2022 forum recommendation (400 mcg folic acid) and the 2024 documentation update (800 mcg methylfolate). Confidence fell below 0.70, triggering the directional prompt. The response correctly identified the newer guidance as the basis for its recommendation while acknowledging the historical context.

Query 7 (follicular phase foods, confidence 0.547): a temporal contradiction was detected between two sources on folate supplementation for MTHFR carriers surfacing alongside an unrelated foods query. The response correctly separated dietary advice (consistent across sources) from supplementation advice (contested), giving clear guidance on foods while flagging the ongoing debate around synthetic folic acid.

Query 10 (8 months trying to conceive, confidence 0.300): the system detected three contradiction signals (two temporal and one version type) among the retrieved candidates. The low confidence score of 0.30 indicated limited certainty about which guidance was definitively superseded, and the response reflected this by offering a structured escalation protocol while clearly flagging that guidelines in this area are evolving.

6. Logging and Source Citation

JSON Log Structure

Every query writes a structured entry to logs/query_log.json containing: ISO 8601 timestamp, raw query string, count of candidates retrieved (always 30), count of candidates after deduplication, a boolean for whether contradictions were detected, a list of contradiction type strings, contradiction confidence score, a list of top-3 source metadata objects (type, title, date), a response generated flag, and latency in milliseconds.

Source Citations

Every generated response appends a formatted source list drawn directly from the top-3 items. Contradiction units render both chunks with a warning marker. The citation format includes source type, document title, version (for documentation), date, and forum engagement metrics (vote count and accepted answer status) where applicable.

UI Presentation

In the Streamlit interface, sources are displayed as color-coded pill chips below the response body: teal for documentation, gold for blog, grey for forum. Chips flagged as contradictions receive an additional amber highlight ring. Each chip is backed by an expandable panel that reveals the full text of the retrieved chunk, giving users transparency into exactly what content drove the generated answer.

7. Performance Analysis

Contradiction Detection

Of the 10 example queries, 3 triggered contradiction handling, which is a rate of 30%. The three queries were: MTHFR folic acid dosing (temporal, confidence 0.66), follicular phase foods (temporal, confidence 0.55), and 8 months trying to conceive (mixed temporal and version, confidence 0.30). Average contradiction confidence across the three detected cases was 0.501, which means all three fell into the directional (low to moderate confidence) prompt path rather than the directive (high confidence) path.

Across all 10 queries, the contradiction scanner identified 17 contradiction signals in total: 12 of type temporal and 5 of type version. Many of these were detected in queries that ultimately did not surface a contradiction in the final top-3, because the conflicting chunks were not both ranked highly enough by the reranker to warrant inclusion.

Deduplication Effectiveness

Average deduplication reduced candidates from 30 to approximately 17, representing a 40.7% reduction. The most productive deduplication occurred on queries touching popular topics with many overlapping source chunks (biomarker sync: 20 remaining; follicular phase: 17 remaining). Deduplication directly reduces the context size passed to both the contradiction scanner and the reranker, lowering token cost and improving signal-to-noise ratio.

Source Distribution in Top-3

Across all 10 queries, 33 total source slots were filled in the top-3 (some queries had 4 sources due to contradiction units expanding to 2 chunks). Of those slots, documentation appeared in 9 instances (27.3%), blogs in 12 instances (36.4%), and forums in 12 instances (36.4%). The relatively high representation of blogs and forums reflects the deliberate balance in the dataset (30 blog chunks, 20 forum chunks) and the per-collection retrieval cap, which prevents documentation from dominating despite its higher credibility score.

Latency

Average query latency across the 10 examples was 41,866 ms (approximately 42 seconds). The fastest query was the conception timing question at 30,425 ms; the slowest was the follicular phase foods query

at 60,383 ms, which incurred additional LLM calls due to multiple contradiction units requiring evaluation. The dominant latency contributors are the embedding API call, the two LLM calls for contradiction detection and reranking, and the generation call, all of which are sequential network round-trips to the Gemini API.

What Worked Well

The contradiction detection pipeline reliably surfaced the deliberate MTHFR and CoQ10 contradictions across multiple queries that touched those topics. RRF fusion produced consistently diverse top-3 results that mixed documentation, forum, and blog sources. Deduplication was effective at removing semantically redundant chunks before the expensive LLM stages. The parent-child injection strategy for documentation ensured that retrieved child chunks carried sufficient context for accurate generation.

Areas for Improvement

Latency is the most significant weakness. Moving from sequential to parallel LLM calls for contradiction detection and reranking would reduce total pipeline time by an estimated 30 to 40 percent. The contradiction confidence formula uses a simple mean of three hand-crafted components; a learned scoring model calibrated against human-labeled contradiction pairs would likely produce better-calibrated confidence values. The current system also lacks a fallback for when the LLM reranker returns malformed JSON, which would cause a silent failure.

8. Ten Example Queries and Responses

Query 1: I have MTHFR C677T variant and how much folic acid should I take?
Contradiction detected: Yes (temporal, confidence 0.66)
Sources: Documentation v2.1 (2024-01-10), Forum (2022-08-15), Blog (2024-03-15), Documentation v2.1 (2024-02-01)
Summary: The system identified a direct conflict between the 2022 recommendation of 400 mcg synthetic folic acid and the 2024 clinical guideline calling for 800 mcg active methylfolate. The response directed the user to switch to methylfolate and explained why synthetic folic acid is contraindicated for this genotype.

Query 2: What CoQ10 dosage is recommended for egg quality?
Contradiction detected: No
Sources: Forum (2022-11-08), Blog (2022-06-10), Documentation v2.1 (2024-02-15)
Summary: The reranker selected the 2024 documentation as the most credible source and anchored the response on the updated 400 to 600 mg daily recommendation for women over 35, noting the prior 200 mg guidance for context. Latency was the highest of all queries at 52,342 ms.

Query 3: How do I sync my biomarker data with the FertiliQ app?
Contradiction detected: No
Sources: Documentation (2024-04-01), Forum (2024-07-03), Forum (2024-09-05)

Summary: The response combined the API documentation with two high-voted forum threads to give a practical step-by-step guide covering LabCorp auto-sync, manual PDF upload, and Oura Ring token renewal. Deduplication was least aggressive here, with 20 of 30 candidates retained.

Query 4: What does a low AMH level mean for my fertility?
Contradiction detected: No
Sources: Blog (2024-05-20), Documentation v2.1 (2024-02-15), Blog (2024-05-20)
Summary: The system correctly identified that low AMH does not mean infertility and provided quantified context (levels as low as 1 to 3 pmol/L have resulted in conception). The response included actionable guidance on vitamin D correction and DHEA supplementation for diminished ovarian reserve.

Query 5: When is the best time to have sex to maximize conception chances?
Contradiction detected: No
Sources: Blog (2024-09-01), Forum (2024-06-10), Documentation v2.1 (2024-03-20)
Summary: The response synthesized sperm survival data (3 to 5 days), egg viability (12 to 24 hours), and a practical every-other-day schedule keyed to LH surge detection. This query had the lowest latency of all ten at 30,425 ms.

Query 6: My vitamin D is 18 ng/mL, so what should I do?
Contradiction detected: No
Sources: Forum (2024-03-15), Documentation v2.1 (2024-02-15), Blog (2024-06-05)
Summary: The response provided a numbered protocol: 5,000 IU vitamin D3 daily, 100 mcg K2 alongside it, administration with the fattiest meal of the day, and retesting at 12 weeks. The system correctly identified 18 ng/mL as well below the fertility-optimal target of 50 to 70 ng/mL.

Query 7: What foods should I eat during the follicular phase?
Contradiction detected: Yes (temporal, confidence 0.55)
Sources: Forum (2024-02-28), Documentation v2.1 (2024-03-01), Forum (2022-08-15), Blog (2024-03-15)
Summary: The system correctly separated food guidance (leafy greens, eggs, wild salmon — all consistent across sources) from supplementation guidance (contested for MTHFR carriers). The response gave a concrete dietary plan while explicitly flagging that folate supplementation recommendations are evolving. Highest latency at 60,383 ms.

Query 8: My partner has low sperm motility and what supplements help?
Contradiction detected: No
Sources: Blog (2024-07-15), Forum (2024-04-02), Blog (2023-11-20)
Summary: The response identified CoQ10 ubiquinol (300 to 400 mg daily), zinc picolinate (25 to 30 mg daily), and selenium (100 to 200 mcg daily) as an evidence-based protocol, explaining the mechanism of each. The 72-day sperm regeneration cycle was cited to frame the 3-month commitment needed to see results.

Query 9: Why does my genetic report say I have high oxidative stress risk?
Contradiction detected: No

Sources: Forum (2023-07-14), Blog (2023-11-20), Blog (2022-06-10)
Summary: The response explained the role of GPX1, SOD2, and GSTM1 variants in antioxidant defense, connected mitochondrial reactive oxygen species to DNA fragmentation index in sperm and oocytes, and placed the genetic risk in practical clinical context. No contradictions were detected because all three sources were internally consistent on the biochemistry.

Query 10: I have been trying for 8 months, what does FertiliQ recommend I check next?
Contradiction detected: Yes (temporal and version, confidence 0.30)
Sources: Documentation v2.1 (2024-03-20), Forum (2024-08-20), Forum (2024-12-03), Blog (2024-09-01)
Summary: The low-confidence contradiction triggered a directional response rather than a directive one. The system recommended checking homocysteine levels and investigating MTHFR variants as the primary next step, while also recommending a shift to multi-marker ovulation tracking. The response clearly communicated that guidelines in this area are still evolving.

9. Real-World Extension

This architecture maps directly onto real medical RAG deployments with only a change in data source rather than pipeline structure. In a clinical setting, the three collections would be replaced by peer-reviewed clinical guidelines (equivalent to documentation with the highest credibility weight), patient community forums such as PatientsLikeMe or disease-specific subreddits (equivalent to forum posts, with vote and endorsement metadata), and systematic review summaries or medical journalism (equivalent to blogs). The contradiction detection and confidence scoring become considerably more consequential in this context: a temporal contradiction between a 2019 chemotherapy dosing guideline and a 2024 update is not a design exercise. It is a patient safety issue. The same pipeline components apply, but the credibility hierarchy would be refined to distinguish, for instance, between FDA guidance, institutional clinical protocols, and individual physician commentary. Retrieval would likely benefit from date-range filtering to prevent significantly outdated sources from entering the candidate pool at all. The source citation and expandable chunk transparency features, already present in FertiliQ, would become a regulatory requirement in many jurisdictions, providing the audit trail needed to verify that generated recommendations are grounded in specific, retrievable sources.